

VIMAL TN

Barcelona, Spain | +34 661171351 | ar.vimaltn@gmail.com | [Portfolio Link](#)

ARCHITECT & AI TECHNOLOGIST | COMPUTER VISION, GENERATIVE AI & GEOSPATIAL INTELLIGENCE FOR THE BUILT ENVIRONMENT

SUMMARY

Architect turned AI practitioner with 6+ years of experience across residential design, construction documentation, and multidisciplinary coordination. Recently completed a Masters in AI for Architecture and Business Innovation at IAAC, Barcelona, building applied AI systems spanning computer vision, generative image workflows, geospatial simulation, and machine-learning decision support.

Combines a practitioner's understanding of how buildings are designed and delivered with hands-on AI development, applying machine learning, computer vision, and generative workflows to real-world challenges in architecture, construction, and cities.

SKILLS

AI & Machine Learning

Python, PyTorch, OpenCV, Computer vision, Image segmentation (SAM), Machine learning, Regression models, Stable Diffusion, generative AI, LLMs

Geospatial & Urban Data

GIS, geospatial data & mapping, Leaflet, FastAPI, Streamlit

Architectural Design & BIM

Revit, AutoCAD, SketchUp, Rhino, BIM modelling, Drawing production

Computational & Digital Tools

Grasshopper, Dynamo, N8N, ROS, Arduino, ESP32, sensor integration

Coordination & Documentation

Multidisciplinary coordination, construction documentation, technical drawings, detailing, quality checks

Visualization

Lumion, Enscape, Photoshop, Illustrator, AI image & video generation

EXPERIENCE

Architect | Partner – CREV Architects | Jun 2020 – Apr 2025

- Developed 3D architectural models and detailed construction drawings for residential projects
- Translated design concepts into coordinated, execution-ready documentation
- Worked with clients and contractors to ensure alignment between design intent and execution
- Managed multiple projects with consistent documentation quality and delivery timelines
- Produced visualizations to communicate spatial and technical solutions effectively

- Successfully designed and delivered around 20 residential projects in 5 years with a project value of around €20 million

Junior Architect | AA Architects | Dec 2018 – Mar 2020

- Prepared 2D drawings and supported architectural design development
- Assisted in 3D modelling and early-stage spatial coordination
- Contributed to construction documentation and approval drawings
- Coordinated design inputs with internal teams for consistency and clarity

| PROJECTS

OASIS | AI Platform for Urban Heat Adaptation | Best Project Award | IAAC | 2026

- Co-created an AI platform helping cities and buildings adapt to extreme heat, working across three scales: urban zone, building, and unit
- Mapped a per-building Heat Vulnerability Index for Barcelona by combining microclimate simulation (UTCI, Infrared SDK), satellite land-surface temperature, building exposure, and social vulnerability data, with LLM-suggested urban interventions and what-if scenarios
- Assessed indoor overheating from uploaded IFC models using health-weighted KPIs (WBGT, nocturnal recovery, age-weighted overheating hours), ranking passive retrofit strategies visualised on the 3D model

Chronoscape | Memory-Augmented Urban Reconstruction | IAAC | 2025–26

- Built an AI image-generation workflow that reconstructs urban environments from memory and lived experience as inputs, supporting urban regeneration design
- Combined conventional urban datasets with oral histories and subjective narratives so memory-augmented models recover spatial relationships and atmospheres that image-only pipelines flatten out

RE:PAIR | Perception-Driven Closed-Loop Robotic Repair | IAAC | 2026

- Developed a closed-loop robotic system that detects surface cracks, repairs them using robotic 3D printing, and autonomously verifies the result
- Trained a computer-vision crack-detection model on a 4,000+ image dataset (Roboflow), with segmentation, masking, and centreline extraction feeding the repair toolpath
- Structured the perception logic and decision flow within a ROS node architecture

Plug and Grow (KATS) | AI Decision Support for Urban Rooftop Farming | IAAC | 2026

- Co-developed an AI dashboard for urban rooftop farming fusing ANN (irrigation optimisation), SVM (disease detection), and Random Forest (system health) models through a weighted decision engine
- Implemented human-in-the-loop feedback so user approvals and corrections rebalance model weights and trigger retraining

- Grounded in Barcelona's 1,764 ha of rooftop farming potential; delivered as a Streamlit app with a digital-twin rooftop map and AI assistant

AI for All | “The Ground Beneath Growth” Interactive GIS Installation | IAAC | 2026

- Built an interactive installation on the Llobregat Delta and the proposed Barcelona–El Prat third runway, letting visitors explore 40 years of change and vote on the delta's future
- Streamed real geospatial data through a FastAPI + Leaflet engine: historical aerial orthophotos (IGN), land cover (GLC_FCS30D), surface temperature (MODIS), live flights (OpenSky), and biodiversity records (GBIF)
- Synchronised a tablet controller with a calibrated ceiling projection over a laser-engraved site model using Server-Sent Events

Social Housing Thesis (Urban Scale)

- Conducted site analysis, density studies, and user behaviour mapping
- Developed housing systems responding to accessibility and spatial efficiency
- Produced masterplans, 3D models, and detailed layouts
- Explored design approaches for resilient and adaptive urban environments

EDUCATION

Masters in AI for Architecture and Business Innovation | IAAC, Barcelona | Sep 2025 – Jul 2026

- The intersection of AI and urban technologies, exploring how data and computation can drive innovation in cities.
- The application of AI in construction and materials, focusing on how intelligent systems are reshaping design, fabrication, and building processes.
- How AI can be harnessed to design adaptive, resilient, and regenerative environments.

Strategic Innovation & Business Development | EADA Business School | Barcelona | Sep 2025 – Jul 2026

- Develop new business models within architecture, engineering, and construction, the Master in AI for Architecture and Business Innovation operates within a robust entrepreneurial framework.

Bachelors In Architecture | Anna university, India | Apr 2013 – June 2018